

Tool support for software maintenance (Scenario #4)

We kindly request your input to explore a scenario that illustrates how tools can be used to notify stakeholders about a project's quality. This questionnaire is organized in three sections. The first section is related to the respondent's consent, and briefly describes the objective of this study and how the collected data will be used. The next section is related to the respondent's profile (e.g., education level and experience). The third section presents the scenario and questions related to it.

* Indicates required question

Survey Consent Statement

1. I hereby give my voluntary and informed consent to participate in the survey titled "Tool support for software maintenance."

*

Purpose: I understand that the purpose of this survey is to collect the perceptions of software practitioners regarding the use of tools for the management of project issues (i.e., issues in issue tracking systems).

Data Usage: I am aware that the data collected through this survey will be used solely for research purposes.

Data Retention: I understand that the data I provide will be stored in an anonymized manner and for a non-defined period of time. This means that my individual responses will not be identifiable, and there is no specified time limit for the retention of this data.

Right to Withdraw: I understand that I have the right to stop the survey at any time and request the withdrawal of my answers. My decision to participate in this survey is entirely voluntary, and I may exercise this right without any negative consequences.

By checking the box below, I confirm that I have read and understood the information provided in this consent statement, and I freely consent to participate in the survey.

Check all that apply.

☐ I confirm

Respondent profile

In this section, we invite you to provide some information related to your education and experience in software development.

2. Highest completed academic degree: *

Mark only one oval.

- ☐ Bachelor
- ☐ Master
- ☐ PhD
- ☐ Other: _____

3. Years of experience in software development in industry: *

Mark only one oval.

- ☐ less than 1 year
- ☐ 1 - 5 years
- ☐ 5 - 10 years
- ☐ more than 10 years

4. Current role: *

Mark only one oval.

- ☐ Software Architect/Designer
- ☐ Software Developer
- ☐ Project Manager
- ☐ Quality Assurance
- ☐ Other: _____

A Travis CI job that notifies stakeholders considering quality metrics calculated by SonarCloud.

In this scenario, Travis CI is used as continuous integration manager. It is integrated with SonarCloud, a tool that analyzes source code to calculate metrics (e.g., the amount of code smells). Within Travis CI, a python script is used as a bot

to collect the results provided by SonarCloud and notifies the stakeholders in a mailing list.

See an example of this scenario below.



5. I consider this scenario **useful** for supporting the notification of stakeholders about a project's quality.

Mark only one oval.

1	2	3	4	5		
Stro	☐	☐	☐	☐	☐	Strongly Agree

6. *When* would this type of notifications be **useful**? For example, after each CI cycle, or if a criterion is met (e.g., the amount of code smells exceeds a threshold). Please justify your answer.

7. *What information* should be provided to **enhance** the communication about a project's overall quality (e.g., amount of code smells, files that contains smells)? Please justify your answer.

8. Is there any other *medium* that you would choose to **receive** the notification instead of a mailing list? Please justify your answer.

9. Is there *any other opinion or comment* about this scenario that you would like to provide?

Conclusion

We sincerely appreciate the time you have dedicated to answering our questionnaire. Your responses will greatly support the understanding and evolution of approaches for tool support in software maintenance.

We kindly invite you to participate in a follow-up interview where we can further investigate your perceptions of tool support for software maintenance and analyze other aspects of the scenario.

10. Would you like to participate in a follow-up interview? *

Mark only one oval.

☐ Yes

☐ No

11. If yes, please provide your e-mail for us to contact you.

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